

FOURTH SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION, APRIL 2022

Mathematics

MAT 4B 04—THEORY OF EQUATION, MATRICES AND VECTOR CALCULUS

(2014—2018 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes**Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 4B 04—THEORY OF EQUATION, MATRICES AND VECTOR CALCULUS

(Multiple Choice Questions for SDE Candidates)

1. If $A = \begin{bmatrix} 2 & 0 \\ 0 & 8 \end{bmatrix}$ then $\rho(A)$ is :
- (A) 0. (B) 1.
(C) 2. (D) n .
2. The points $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ are collinear if and only if the rank of the matrix $\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$ is :
- (A) < 3 . (B) ≤ 3 .
(C) > 3 . (D) ≥ 3 .
3. If a matrix A can be reduced to the normal form $\begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$ by using elementary operations, then $\rho(A)$ is :
- (A) 4. (B) 3.
(C) 2. (D) 1.
4. $\text{Rank}(AA') = \underline{\hspace{2cm}}$.
- (A) $\text{rank } A$. (B) $\text{rank } A'$.
(C) 1. (D) None.
5. For a system of m linear equations in unknowns, Cramer's rule is applicable, when ?
- (A) $m = n$.
(B) $m \neq n$.
(C) $m = n$ and the coefficient matrix is non-singular.
(D) $m = n$ and the coefficient matrix is singular.

6. The system of equations $x + 2y + z = 9$ can be expressed as :
 $2x + y + 3z = 1$.

(A) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

(B) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} 9 \\ 7 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

(C) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \end{bmatrix}$.

(D) None.

7. If A is a square matrix of order n and λ is a scalar, then the characteristic polynomial of A is obtained by expanding the determinant.

(A) $|\lambda A|$.

(B) $\lambda |A|$.

(C) $|\lambda A - I_n|$.

(D) $|A - \lambda I_n|$.

8. The scalar λ is a characteristic root of the matrix A if :

(A) $(A - \lambda I)$ is non singular.

(B) $(A - \lambda I)$ is singular.

(C) A is non singular.

(D) A is singular.

9. If eigen value of matrix A is λ , then eigen value of $p^{-1}AP$ is :

(A) 1.

(B) λ .

(C) $1/\lambda$.

(D) 0.

10. A polynomial equation whose roots are 3 times those of the equation $2x^3 - 5x^2 + 7 = 0$ is :

(A) $3x^3 - 15x^2 + 21 = 0$.

(B) $2x^3 - 15x^2 + 189 = 0$.

(C) $2x^3 + 15x^2 - 189 = 0$.

(D) None.

11. If α, β, γ are the roots of $x^3 + px^2 + qx + r = 0$ then $\alpha\beta + \beta\gamma + \gamma\alpha$ equals .

(A) $-\frac{p}{q}$.

(B) p .

(C) q .

(D) $-q$.

12. If I is a unit matrix of order n , then its rank is equal to :

(A) 1.

(B) n .

(C) Less than n .

(D) Greater than n .

13. The rank of the matrix $\begin{bmatrix} 1 & 3 & 4 & 2 \\ 0 & 2 & 1 & 4 \\ 0 & 0 & 2 & 0 \end{bmatrix}$ is :

(A) 3.

(B) 4×3 .

(C) 2.

(D) 1.

14. Rank of the matrix $A = \begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 9 \\ -1 & -3 & -4 & -3 \end{bmatrix}$ is :

(A) 1.

(B) 2.

(C) 3.

(D) 4.

15. If a matrix A has a nonzero minor of order r , then :

(A) $\rho(A) = r$.

(B) $\rho(A) \geq r$.

(C) $\rho(A) < r$.

(D) $\rho(A) \leq r$.

16. Which of the following is false ?

(A) $\rho(A+B) \leq \rho(A) + \rho(B)$.

(B) $\rho(A') = \rho(A)$.

(C) $\rho(A+B) = \rho(A) + \rho(B) - 4$, if A and B are matrices of rank 4.

(D) $\rho(A-B) \leq \rho(A)\rho(B)$.

17. A system of m non-homogeneous linear equations $Ax = B$ in n unknown is consistent if :

(A) $m = n$.

(B) $m \neq n$.

(C) $\rho(A) \neq \rho([A, B])$

(D) $\rho(A) = \rho([A, B])$

18. A system of m homogeneous linear equations $Ax = 0$ in n unknown has only trivial solution if :

(A) $m = n$.

(B) $m \neq n$.

(C) $\rho(A) = m$.

(D) $\rho(A) = n$.

19. The characteristic roots of skew-Hermitian matrix are either :

(A) Real or zero.

(B) Real or non-zero.

(C) Pure imaginary or zero.

(D) Pure imaginary or non-zero.

20. The matrix $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ satisfies the equation :

(A) $A^2 + 5A + 7I = 0$.

(B) $A^2 + 5A - 7I = 0$.

(C) $A^2 - 5A - 7I = 0$.

(D) $A^2 - 5A + 7I = 0$.

**FOURTH SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
APRIL 2022**

Mathematics

MAT 4B 04—THEORY OF EQUATION, MATRICES AND VECTOR CALCULUS

(2014—2018 Admissions)

Time : Three Hours

Maximum : 80 Marks

Part A (Objective Type)

Answer all twelve questions.

Each question carries 1 mark.

1. State the fundamental theorem of theory of equations.
2. If α, β, γ are the roots of the equation $ax^3 + bx^2 + cx + d = 0$, write the equation whose roots are $-\alpha, -\beta, -\gamma$.
3. Find the number of real roots of $x^4 - 1 = 0$.
4. Write the standard form of a cubic equation.
5. Find the rank of $\begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$.
6. If A and B are non-singular square matrices of order 5, find the rank of AB.
7. Find the number of solutions of the equation $x + 2y = 3$.
8. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 4 & 2 \\ a & 6 & b \end{bmatrix}$ and the system of homogeneous linear equations $AX = 0$ has a non-zero solution, find the value of b .
9. Find the characteristic roots of $\begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$.
10. Find the parametric equations of the line through the point $(3, -4, -1)$ parallel to the vector $\mathbf{i} + \mathbf{j} + \mathbf{k}$.

Turn over

11. Find the angle between the planes $x + y = 1$, $2x + y - 2z = 2$.
12. Find the unit tangent vector to the curve $\mathbf{r}(t) = (\cos t) \mathbf{i} + (\sin t) \mathbf{j}$.

(12 × 1 = 12 marks)

Part B (Short Answer Type)

*Answer any nine questions.
Each question carries 2 marks.*

13. If α, β, γ are the roots of the equation $2x^3 + x^2 - 2x - 1 = 0$, find the value of $\alpha + \beta + \gamma$.
14. If α, β, γ are the roots of the equation $2x^3 + 3x^2 - x - 1 = 0$, find the equation whose roots are $\frac{1}{2\alpha}, \frac{1}{2\beta}, \frac{1}{2\gamma}$.
15. Show that the equation $x^4 + 4x^2 + 5x - 6 = 0$ has exactly one positive root.
16. Show that the rank of a matrix, every element of which is unity is 1.
17. Find the normal form of $\begin{bmatrix} 1 & 2 & -3 \\ 2 & 5 & -4 \end{bmatrix}$.
18. Find the values of λ so that the system of equations $\lambda x + y = 0$, $x + \lambda y = 0$ has zero solution only.
19. Prove that the characteristic roots of triangular matrix are the same as its diagonal elements.
20. Show that if λ is a characteristic root of a matrix A , then $\lambda + k$ is a characteristic root of the matrix $A + kI$.
21. Find the spherical co-ordinate equation for the sphere $x^2 + y^2 + (z - 1)^2 = 1$.
22. If $\mathbf{r}(t) = (3 \cos t) \mathbf{i} + (3 \sin t) \mathbf{j} + t^2 \mathbf{k}$ is the position vector of a particle in space at time t , at what times, if any, are the body's velocity and acceleration orthogonal?
23. If $\mathbf{u}$ is a differentiable vector function of t of constant magnitude, prove that $\mathbf{u} \cdot \frac{d\mathbf{u}}{dt} = 0$.
24. Show that the curvature of a straight line is zero.

(9 × 2 = 18 marks)

Part C (Short Essay Type)

*Answer any six questions.
Each question carries 5 marks.*

25. Solve $4x^3 - 24x^2 + 23x + 18 = 0$, given that the roots are in A.P.
26. If α, β, γ are the roots of the equation $x^3 + 3x^2 + 6x + 1 = 0$, find the value of $(\alpha^2 + 1)(\beta^2 + 1)(\gamma^2 + 1)$.
27. Obtain the real root of the equation $x^3 - 15x = 126$ by Cardan's method.
28. Reducing to the normal form, find the rank of $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$.
29. If $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix}$, find non-singular matrices P and Q such that PAQ is in normal form.
30. Test for consistency and solve the system of equations :
- $$5x + 3y + 7z - 4 = 0$$
- $$3x + 26y + 2z - 9 = 0$$
- $$7x + 2y + 10z - 5 = 0.$$
31. If A is a non-singular matrix, prove that the eigenvalues of A^{-1} are the reciprocals of the eigenvalues of A.
32. Find the distance from the point (1, 1, 5) to the line $x = 1 + t, y = 3 - t, z = 2t$.
33. Obtain the curvature of a circle of radius a .

(6 × 5 = 30 marks)

Turn over

Part D (Essay Type)

*Answer any **two** questions.
Each question carries 10 marks.*

34. Solve $6x^5 + 11x^4 - 33x^3 - 33x^2 + 11x + 6 = 0$.

35. Find the characteristic roots and the corresponding characteristic vectors for the matrix

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}.$$

36. Find the binormal vector and torsion for the space curve $\mathbf{r}(t) = (3 \sin t) \mathbf{i} + (3 \cos t) \mathbf{j} + 4t\mathbf{k}$.

(2 × 10 = 20 marks)